

REPORT

:: Memory & State Management

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Technical Note

AI Personal Assistant with Memory & State Management

Introduction

In this project, I developed a simple AI Personal Assistant that can remember information provided by the user. The purpose of this project is to understand how memory works in AI systems and how an assistant can use previous information to provide better responses.

Normally, a chatbot forgets everything when the program is closed. To solve this problem, I implemented both Short-Term Memory and Long-Term Memory. Short Term Memory helps the assistant remember the current conversation, while Long Term Memory stores important information for future use.

Project Overview

The project is based on the idea of a personal assistant that can remember user preferences. For example, if a user says, "My favorite color is black," the assistant stores this information.

Later, when the user asks about their favorite color, the assistant can retrieve the stored information and answer correctly.

This makes the chatbot more useful and creates a better user experience.

Short Term Memory

Short-Term Memory is used to keep track of the current conversation. It stores recent messages exchanged between the user and the assistant.

For example:

- User: My name is Sania.
- Assistant: Nice to meet you, Sania.

If the user later asks, "What is my name?" during the same session, the assistant can answer correctly because the information is still available in memory.

However, this memory is temporary and is cleared when the program is closed.

Long Term Memory

Long-Term Memory is used to save important information permanently. For this purpose, I used ChromaDB, which is a vector database commonly used in AI applications.

The assistant stores useful user information such as:

- Name
- Favorite color
- Favorite city
- Interests and preferences

Even after restarting the application, the assistant can retrieve this information and use it in future conversations.

Tools and Technologies Used

The following tools were used in this project:

- Python
- ChromaDB
- Visual Studio Code (VS Code)
- Virtual Environment (venv)

Python was used to build the application, while ChromaDB was used to store and retrieve user memories.

Working Process

The working process of the system is simple:

1. The user enters a message.
2. The assistant analyzes the message.
3. Important information is stored in memory.
4. The assistant searches memory whenever needed.
5. The most relevant information is returned to the user.

This process allows the assistant to remember facts and provide personalized responses.

Challenges Faced

While working on this project, I learned how memory management works in AI systems. Setting up the environment and understanding how vector databases store information were the main challenges. After learning the basic concepts, it became easier to implement the memory system.

Conclusion

This project helped me understand the concept of Memory and State Management in AI applications. By combining Short-Term Memory and Long-Term Memory, the assistant can remember user information and provide more intelligent responses. This project can be further improved by adding a graphical interface, voice support, and integration with advanced AI models.

Memory architecture diagram

